

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and optimized components for a medical device, translating design inputs into tangible engineering specifications and drawings using **Java Spring Boot** and **Python**.
- Assisted in process development activities, analyzing key process inputs and outputs using **statistical methods** and Design of Experiments (DOE) to improve product yield by **15%**.
- Authored technical documents, validation protocols, and reports, including raw material specifications and drawings, ensuring compliance with **ISO standards** and FDA regulations.
- Designed and implemented new engineering test methods and simulated use test methods, collaborating with R&D; technicians and validating over **5 new methods**.
- Performed hands-on laboratory testing, including engineering evaluations and competitor analyses, contributing to the successful verification and validation of **3 new product features**.
- Supported cross-functional teams across R&D;, Quality, and Operations to ensure system compliance with internal and external guidelines, resolving over **50** non-conformance issues.

SKILLS

Medical Device Development – Design Control, Process Development, Test Method Development, Verification & Validation (V&V;), IQ/OQ/PQ, Design Inputs/Outputs, Clinical Use Scenarios, International Standards.

Engineering & Analysis – Statistical Methods, Design of Experiments (DOE), Root Cause Analysis, Data Analysis, Regression Analysis, Hypothesis Testing, Time-Series Analysis.

Software & Tools – Python, Java, Spring Boot, SQL, R, RStudio, pandas, NumPy, scikit-learn, Flask, FastAPI, Power BI, Tableau, Power Query, DAX.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux, Shell Scripting.

Documentation & Compliance – Technical Writing, Work Instructions, Standard Operating Procedures (SOPs), cGMP, GLP, Regulatory Affairs, GDPR, Audit Trails.

Project Management & Collaboration – Agile/Scrum, Jira, Confluence, Cross-functional Teams, Mentoring, Technical Training, Change Management.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Medical Device Component Optimization Platform

- Developed a Python-based platform to analyze and optimize key parameters for novel medical device components, integrating data from simulated use testing.

- Implemented automated reporting using **Power BI**, visualizing process inputs, outputs, and yield improvements, providing actionable insights to the R&D; team.
- Created detailed technical documentation and work instructions for component manufacturing processes, ensuring adherence to internal quality standards.

Automated Test Method Validation Framework

- Designed and built a framework using **Java Spring Boot** and **SQL** to automate the validation of engineering test methods for medical device components.
- Integrated with AWS services for data storage and processing, enabling faster test method validation cycles and reducing manual effort by **30%**.
- Authored validation protocols and reports, collaborating with Quality Assurance to meet regulatory requirements for new product introductions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)